

Gamma detection for source localisation in targeted alpha therapy

C. De Sio¹, L. Abu Sabah¹, J. Duan¹, Y. Shi¹, S. Williams¹ and J. Velthuis¹

¹School of Physics, University of Bristol, Bristol, United Kingdom

Abstract. Targeted alpha therapies can deliver a highly localised dose to tumours, but treatment planning and effectiveness depend on whether the radionuclide source remains near the intended site after administration. In the AlphaGlue delivery concept, alpha emitters are applied to a tumour region in a thin glue layer. We present a Geant4 simulation study showing that decay-chain gamma detection can distinguish retained, broadly diffused and locally displaced source states in an AlphaGlue-like geometry. For each of the seven source configurations, 50 independent simulations of 10,000 primary events were generated, and baseline-subtracted gamma-hit profiles were successfully used to reconstruct the broad diffusion fraction with $R^2 = 0.9987$ and a mean absolute error of about one percentage point. Displaced source activity was detected by gamma signal analysis and source position was reconstructed with mean position errors of 16–30 mm. These results support gamma detection as a candidate method for emitter-position monitoring, treatment planning and delivery verification.

1 Introduction

In targeted alpha therapy, the delivered dose is strongly local. A treatment plan therefore depends on the initial activity distribution and its evolution after administration. The AlphaGlue concept proposes suspending alpha emitters in a thin glue layer applied to tissue [1], as illustrated in Fig. 1. The concept has previously been evaluated with Geant4 and Geant4-DNA simulations for dose and DNA-damage modelling [2–7]. Its delivered dose depends on whether source material and progeny remain near the intended deposition site, diffuse into surrounding tissue or are redistributed by biological transport [8,9]. This study investigates whether the accompanying gamma signal can be used as a theranostic observable to monitor emitter position, detect source movement and provide useful information as a function of detection time.

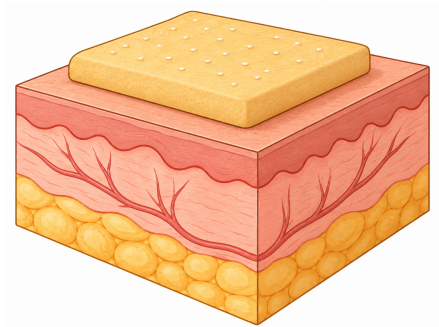


Figure 1: Conceptual illustration of an AlphaGlue applicator placed on tissue.

2 Simulation model

Photon transport was simulated with the Geant4 toolkit [2]. The model places a $40\text{ mm} \times 2\text{ mm} \times 40\text{ mm}$ loaded glue layer on the $+y$ surface of a $20\text{ cm} \times 20\text{ cm} \times 150\text{ cm}$ water phantom. Six detector panels surround the phantom at 20 cm from the water surface. Each photon entry into a detector panel is counted as one hit.

Radionuclides or their progeny may leave the applicator region by diffusing into surrounding tissue or entering the circulation and being transported by blood flow. This migration can produce a broad redistribution through the surrounding tissue, with large variation reported between experiments [8,9]. To test whether these behaviours can be distinguished from source retention, seven source configurations were simulated. The retained-source configuration was used as the baseline. Broad redistribution was represented by cases in which 10%, 50% or 90% of the source was distributed throughout the water volume. The localised-displacement scenario was represented by placing 20%, 40% or 60% of the source in a 5 cm-side water cube centred at $(x, y, z) = (0, 0, 500 \text{ mm})$, spanning $475 \leq z \leq 525 \text{ mm}$.

The geometry is described in Fig. 2, along with example detector displays showing the three clinical scenarios. Each configuration was simulated with 50 independent repetitions of 10,000 primary decays, giving 500,000 simulated decays per configuration and 3.5 million simulated decays across all source states. The repeated runs define the run-to-run uncertainty used in the residual and detectability analysis. The simulated decays are rescaled to a DaRT-like reference activity of $3 \mu\text{Ci}$ (111 kBq), matching the order of a single DaRT seed [10].

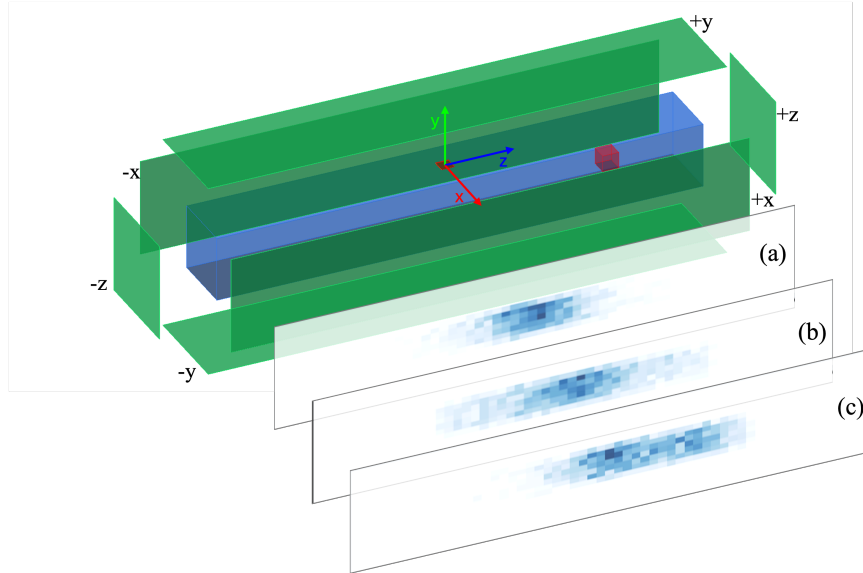


Figure 2: Simulation geometry showing the water phantom (blue), AlphaGlue applicator region (red, centre), surrounding detector panels (green), a displaced source (red, right side) and three offset example x -panel projections: baseline (a), broad redistribution (b), and displaced localisation (c).

3 Baseline subtraction

Each photon entry into a detector panel was recorded as one hit, together with its detector panel and entry position. Detector panels are named according to the side of the phantom on which they are located. The $+x$ and $-x$ panels are the two lateral panels at positive and negative x ; the $+y$ panel faces the AlphaGlue applicator and the $-y$ panel lies on the opposite side. The $+z$ and $-z$ panels close the longitudinal ends of the geometry. These end panels recorded substantially fewer entries than the four side panels and were not included in the present quantitative analysis. A combined x -panel profile is obtained by summing the $+x$ - and $-x$ -panel counts bin by bin and plotting them against the longitudinal coordinate z . Thus, “ x -panel” identifies the detector

location, while z identifies position along the panel. Longitudinal profiles were examined because redistribution changes the spatial pattern of detector hits more strongly than the total number of hits. The total detector-entry rate differed by only 3.8% between the retained baseline and the 90% broad-diffusion case, whereas subtraction of the retained-source profile revealed clear spatial differences (Fig. 3). The retained source defines the reference profile $\bar{h}_p^B(z)$ for detector panel p . Candidate source states are represented by

$$\Delta h_p(z) = h_p(z) - \bar{h}_p^B(z), \quad (1)$$

with the repeated baseline simulations defining the bin-wise noise scale. Each profile was divided by the number of simulated primary decays and rescaled to the initial emitter population corresponding to the $3 \mu\text{Ci}$ reference activity. The plotted residual is the difference between the mean activity-scaled candidate and retained-source profiles. Positive residuals mark detector regions where the candidate produces more gamma entries than the retained-source case, while negative residuals mark depleted regions. In the combined (x)-panel projection, broad diffusion widens the residual pattern, whereas localised displacement produces a shifted excess near the displaced source region. Changes in the balance between the glue-facing and opposite detector panels provide an additional feature for the quantitative analysis.

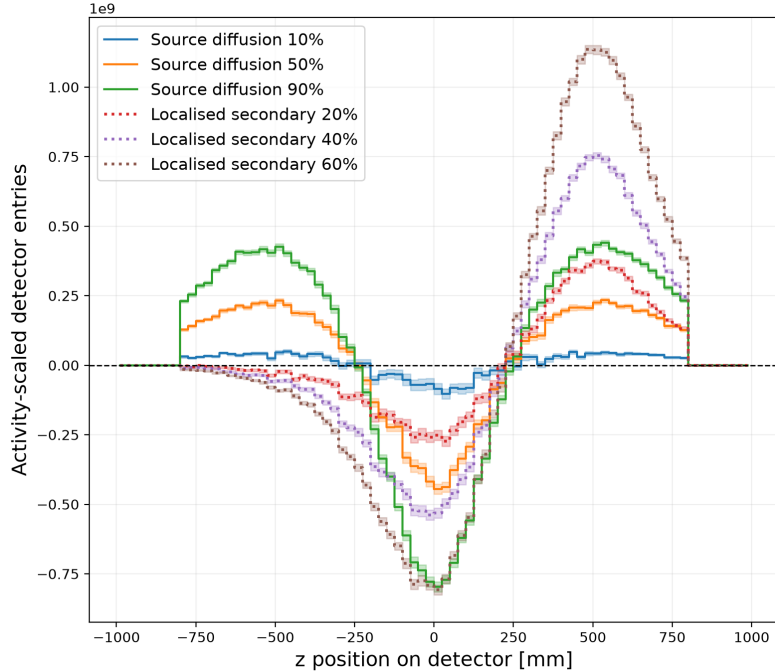


Figure 3: Activity-scaled, baseline-subtracted combined $+x$ and $-x$ detector profiles for broad diffusion and localised displacement. Shaded bands show the propagated standard error of the mean across 50 repetitions.

As an illustration of the general ROI strategy, Fig. 4 compares the full longitudinal profile formed by summing the $+x$ and $-x$ panel counts with a region at $-750 \leq z \leq -550$ mm, where little signal is expected from retained activity. An excess in this region indicates that the detected distribution has spread beyond the applicator footprint. This ROI illustrates the principle on which the quantitative estimators in Section 4 are based: source redistribution is inferred from spatial changes relative to the retained-source reference.

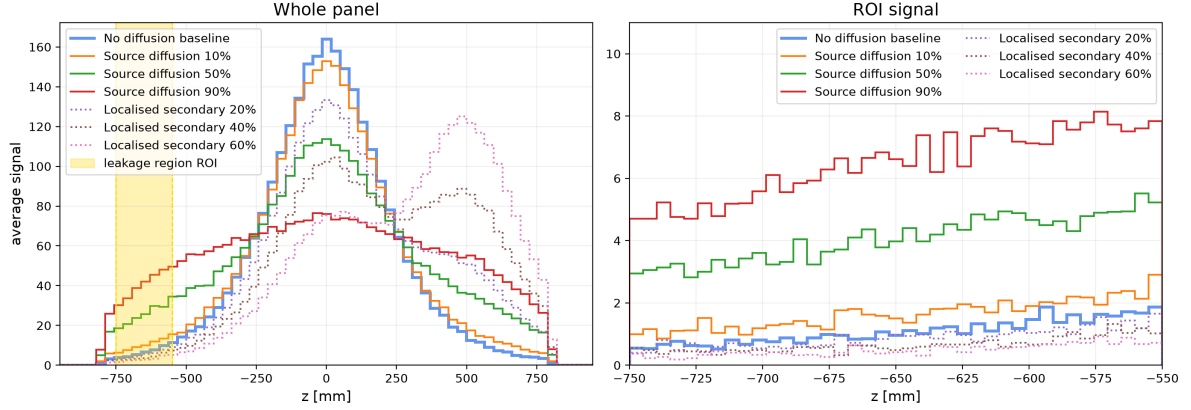


Figure 4: Combined $+x/-x$ longitudinal profiles (left) and an illustrative low-signal ROI (right). The ROI highlights the broad-diffusion excess outside the retained-source region.

4 Signal analysis and source redistribution metrics

The signal analysis addresses four tasks: detecting and locating a compact secondary source, estimating the broadly diffused fraction when no compact source is present, estimating how much signal remains consistent with the applicator region, and determining the observation time required to distinguish each candidate from the retained-source baseline. Together, these tasks determine whether activity has moved, whether the redistribution is broad or localised, where a compact displaced source is located, and how quickly each change becomes detectable. Both broad diffusion and compact displacement can reduce the central signal, so the compact-secondary test is applied first. A negative result enables the broad-diffusion and applicator-retention estimates; a positive result instead triggers source localisation.

4.1 Compact secondary-source detection and localisation

Counts and spatial-profile features from the four side panels were used to distinguish a compact displaced source from retained and broadly redistributed source states. The panel asymmetries were calculated as

$$A_i = \frac{N_{+i} - N_{-i}}{N_{+i} + N_{-i}}, \quad i \in \{x, y\},$$

where N_{+i} and N_{-i} are the total numbers of entries on the opposing side panels. Broad diffusion primarily changes A_y , because activity moves away from the applicator-facing $+y$ panel, while a source displaced along z produces a shifted feature within the longitudinal detector profiles. Compact-source detection used a balanced logistic-regression classifier supplied with standardised panel asymmetries, baseline-subtracted profile bins, positive-residual integrals, peak and centroid positions, longitudinal widths, ROI rates and the total detected rate. Performance was evaluated using leave-one-repetition-out validation.

A displaced-source ROI of $250 \leq z \leq 750$ mm was defined in the combined $+x/-x$ and $+y/-y$ longitudinal profiles. For each view, the ROI signal was obtained by summing the positive candidate-minus-baseline residual over all z bins in this interval. After positive detection, the source position was estimated from the positive-residual centroid separately in the combined x - and y -panel profiles. The final position estimate was the mean of these two centroids.

$$\hat{z}_{\text{sec}} = \frac{\hat{z}_{X\text{sum}} + \hat{z}_{Y\text{sum}}}{2},$$

The reconstructed positions and their repetition-level residuals are reported in Fig. 5.

4.2 Broad-diffusion fraction

Broad diffusion is estimated only after the compact-secondary test is negative. Inputs comprise shape-normalised z -profile bins from the combined x , combined y and combined four-side-panel views; the standard deviation and central-80% interquantile width ($z_{90} - z_{10}$); the central $+y$ fraction; x - and y -panel asymmetries; absolute-residual integrals; and positive-residual centroid and peak-position summaries. These features are standardised and supplied to ridge regression [11]. The ridge regularisation parameter, α , was selected by cross-validation from candidate values between 10^{-4} and 10^4 , balancing agreement with the training data against model complexity. The model is trained only on the retained and 10%, 50% and 90% broad-diffusion states; localised-source cases are excluded. Performance is evaluated with leave-one-repetition-out validation. The resulting predictions are shown in Fig. 6.

4.3 Applicator-retained signal

The applicator-facing observable is the $+y$ -panel count in the central ROI, $|z| \leq 150$ mm, normalised per primary event. Dividing this rate by the retained-source rate gives a baseline-relative proxy for the fraction of signal remaining at the applicator. The combined four-side-panel central ROI is retained as a cross-check. As shown in Fig. 7, both broad diffusion and compact displacement reduce this signal, so an applicator-retained fraction is quoted only after the compact-secondary test has been evaluated.

4.4 Time-window detectability

The same baseline-subtracted profiles and task-specific ROIs are accumulated in increasing observation windows. At each observation window, the mean candidate and baseline profiles are compared bin by bin. The uncertainty is the quadrature sum of their standard errors across the 50 repetitions. A source state is considered detectable when the maximum absolute candidate-minus-baseline residual exceeds three times this combined standard error. The reported detection time is the first cumulative window that crosses this threshold. This procedure tests when the spatial redistribution used by the diffusion and localisation estimators becomes statistically resolvable; the resulting threshold times and an example cumulative profile are presented in Section 5.2.

5 Results

5.1 Signal inference

Using profile-shape features from repeated simulations, the broad diffusion fraction was recovered with $R^2 = 0.9987$ and a mean absolute error of about one percentage point (Fig. 6). The baseline-normalised central $+y$ -panel ROI signal decreased monotonically with broad diffusion, to 0.925 ± 0.017 , 0.618 ± 0.013 and 0.289 ± 0.009 for the 10%, 50% and 90% broad-diffusion cases, respectively (Fig. 7). The ratio measures the central-ROI detector signal relative to the retained-source baseline; it is not a direct measurement of the fraction of activity remaining in the applicator. The compact-source strategy identified the localised displaced source in all 20%, 40% and 60% localised-displacement cases. The reconstructed displaced-source positions were approximately $z = 516$ mm, 528 mm and 530 mm, respectively, compared with the simulated source centre at $z = 500$ mm. Across these localised fractions, mean position errors were 16–30 mm, with the repetition-level residual distributions shown in Fig. 5. The repetition-to-repetition spread decreased from 6.6 mm to 2.1 mm as the displaced-source fraction increased, while the

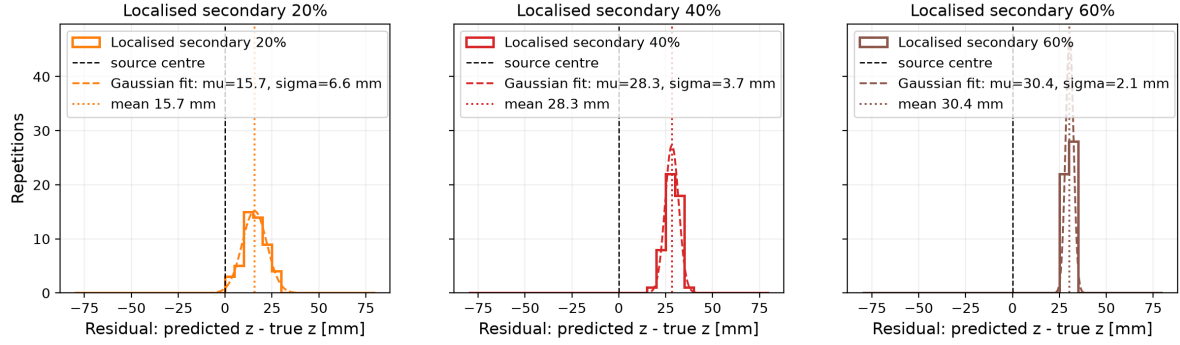


Figure 5: Localisation residuals across 50 repetitions for each displaced-source fraction. The dashed black line marks the true source centre; coloured lines indicate the mean and Gaussian fit.

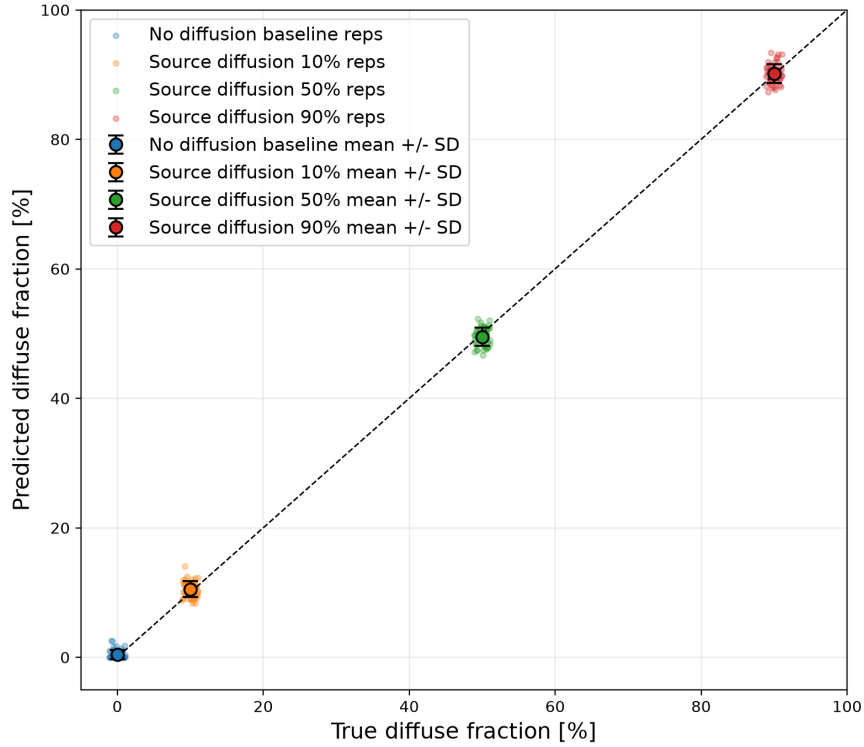


Figure 6: Diffusion fraction inferred from 50 independent repetitions. Small points show individual repetitions and large markers show the mean and standard deviation.

mean estimate remained shifted towards positive z . This indicates increasing precision but a systematic bias in the positive-residual centroid estimator; calibration using simulations at multiple source positions will be required in future work.

5.2 Time dependence

Cumulative time-window profiles show that the baseline-subtracted signal is already visible on hour-scale windows and becomes stronger over longer observation windows. Across the 50 simulation repetitions, detectability was defined as a maximum absolute bin-wise residual

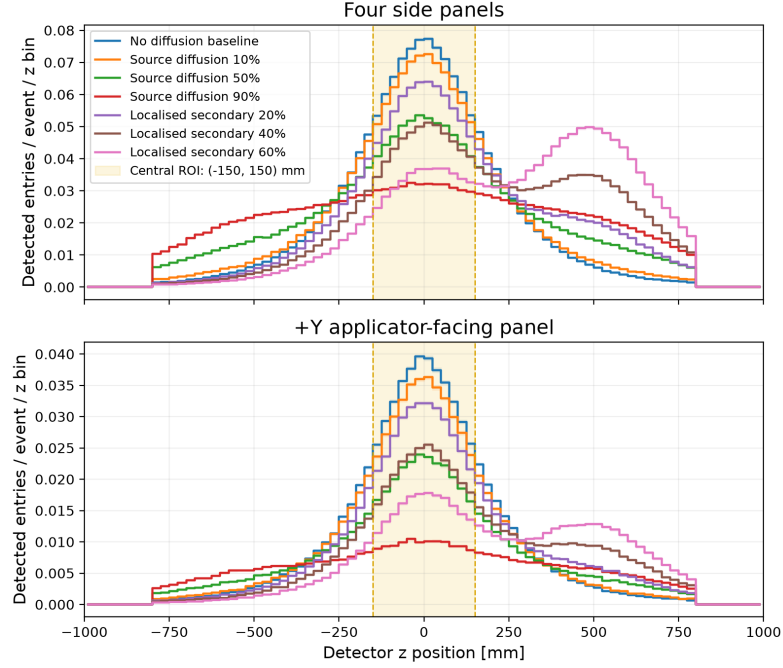


Figure 7: Central $|z| \leq 150$ mm ROI for the four side panels (top) and the applicator-facing $+y$ panel (bottom). The $+y$ ROI rate provides the baseline-relative applicator-retained signal.

Table 1: First 3σ detection times from the time-window analysis of 50 independent repetitions.

Case group	First 3σ detection
10% broad diffusion	about 6 h
50–90% broad diffusion	about 1–2 h
20–60% localised displacement	about 1–2 h

greater than three times the combined candidate and baseline standard error. The 50–90% broad-diffusion and 20–60% localised-displacement cases crossed threshold in about 1–2 hours, while the weaker 10% broad-diffusion case required about 6 hours. A representative time-window profile is shown in Fig. 8. The first threshold-crossing times are summarised in Table 1.

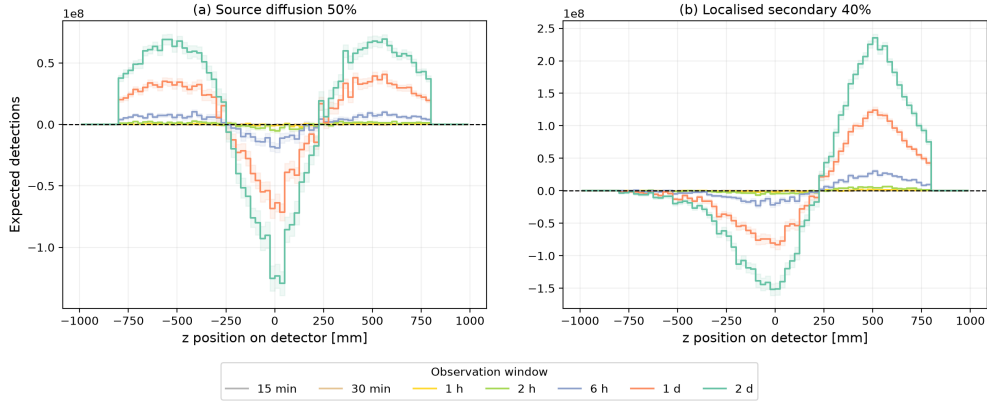


Figure 8: Cumulative combined $+x/-x$ residual profiles for 50% broad diffusion (left) and 40% localised displacement (right). Broad diffusion produces a symmetric redistribution away from the centre, whereas localised displacement produces an asymmetric excess near the displaced-source region. These signatures first exceed the 3σ detection threshold after approximately 2 hours and 1 hour, respectively.

6 Conclusion

Decay-chain gamma detection provides source-location information and broad redistribution estimation. Baseline-subtracted gamma profiles distinguished retained, broadly redistributed and locally displaced source states. Broad-diffusion fraction was reconstructed with $R^2 = 0.9987$ and a mean absolute error of approximately one percentage point. All simulated compact-source cases were detected, with mean reconstructed positions 16–30 mm from the simulated source centre. Depending on the source state, redistribution became detectable after approximately 1–6 hours. Broad diffusion and compact source displacement were treated as independent source states in the present analysis. A combined case, in which diffuse redistribution and a compact displaced source occur together, should be included in the next simulation and inference studies. Further work will also include detector response, background estimation and patient-scale geometries so that activity-normalised profiles can be interpreted as experimentally measurable count rates.

References

- [1] Abu Sabah L *et al.* 2025 AlphaGlue: A novel conceptual delivery method for α therapy *BioMedInformatics* **5** 58
- [2] Agostinelli S *et al.* 2003 Geant4—a simulation toolkit *Nucl. Instrum. Methods Phys. Res. A* **506** 250–303
- [3] Incerti S *et al.* 2010 The Geant4-DNA project *Int. J. Model. Simul. Sci. Comput.* **1** 157–178
- [4] Ballisat L *et al.* 2025 Simulation of cell cycle effects on DNA strand break induction due to α -particles *Phys. Med.* **129** 104871
- [5] De Sio C *et al.* 2025 Targeted alpha therapies using ^{211}At : A Geant4 simulation of dose and DNA damage *Phys. Med.* **129** 104860
- [6] Ballisat L *et al.* 2024 Dose and DNA damage modelling of diffusing alpha-emitters radiation therapy using Geant4 *Phys. Med.* **121** 103367

- [7] Ballisat L *et al.* 2023 In-silico calculations of DNA damage induced by α -particles in the ^{224}Ra DaRT decay chain for a better understanding of the radiobiological effectiveness of this treatment *Phys. Med.* **112** 102626
- [8] Ballisat L *et al.* 2025 Geant4 simulation of the enhanced in-vivo ^{220}Rn spread in DaRT *Phys. Med.* **135** 105005
- [9] Heger G *et al.* 2024 First measurements of radon-220 diffusion in mice tumors *Med. Phys.* **51** 5045–5058
- [10] Arazi L *et al.* 2010 The treatment of solid tumors by alpha emitters released from ^{224}Ra -loaded sources *Phys. Med. Biol.* **55** 1203
- [11] Hoerl A E and Kennard R W 1970 Ridge regression: Biased estimation for nonorthogonal problems *Technometrics* **12** 55–67